Sophia Systems 2006-12-1

CF Operation Check Procedure

■ CF Boot Method

Operating environment:

CF card: SanDisk SDCFB 512MB

File system format: ext2

1. Copy the file system to the CF card.

Use FedoraCore5 to format the CF card and then copy the file system to it. The input commands are shown in red.

[]# /sbin/fdisk /dev/sda

When fdisk starts up, if a new disk is being used, use the “n” command to create a new partition and then use “p” to select “p Basic Area (1-4).”

Command (m for help): n

e extended

p primary partion (1-4)

Use p to enter “1.” Below, enter the first cylinder numerical value and then enter the last cylinder numerical value.

Partition number (1-4): 1

First cylinder (1-1009, default 1): 1

Last cylinder or +size or +sizeM or +sizeK (1-1009, default 1009): 1009

Use the “p” command to check the setting.

Command (m for help): p

Disk /dev/sda: 512 MB, 512483328 bytes

16 heads, 62 sectors/track, 1009 cylinders

Units = cylinders of 992 * 512 = 507904 bytes

Device	Boot	Start	End	Blocks	Id	System
/dev/sda1		1	1009	500433	83	Linux

The numerical value set in “sda” as attached and displayed. At this time, make a memo of the “Blocks” value.

When finished making the setting, enter the “w” command to write the setting. The setting will not actually be applied if the “w” command is not executed. The setting will also not be written if the “q” command is used to exit.

Command (m for help): w

```
The partition table has been altered!

Calling ioctl() to re-read partition table.
SCSI device sda: 1000944 512-byte hdwr sectors (512 MB)
sda: Write Protect is off
sda: assuming drive cache: write through
SCSI device sda: 1000944 512-byte hdwr sectors (512 MB)
sda: Write Protect is off
sda: assuming drive cache: write through
Syncing disks.

[]#  /sbin/mkfs  -t  ext2  /dev/sda1  500433
mke2fs 1.38 (30-Jun-2005)
Filesystem label=
OS type: Linux
Block size=1024 (log=0)
Fragment size=1024 (log=0)
125488 inodes, 500433 blocks
25021 blocks (5.00%) reserved for the super user
First data block=1
Maximum filesystem blocks=67633152
62 block groups
8192 blocks per group, 8192 fragments per group
2024 inodes per group
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729, 204801, 221185, 401409

Writing inode tables: done
Writing superblocks and filesystem accounting information: done

This filesystem will be automatically checked every 22 mounts or
180 days, whichever comes first.  Use tune2fs -c or -i to override.

[]#  mkdir  /mnt/cf/
[]#  mount  /dev/sda1  /mnt/cf/
[]#  cp  -rf  /opt/target/rootfs-xscale-060924-2/*  /mnt/cf/
     Copy the file system to the CF card.
[]#  umount  /mnt/cf/
```

2. Kernel Creation

The .config of “5.2.1 Kernel Configuration” of the Handbook was changed. Accompanying the change, please create zImage.

①. Change CONFIG_CMDLINE.

CONFIG_CMDLINE="console=ttyS0,115200 mem=64M@0xa0000000 root=/dev/mtdblock1 rootfstype=jffs2"

↓ Change

CONFIG_CMDLINE="console=ttyS0,115200 mem=64M@0xa0000000 root=/dev/hda1 rootfstype=ext2"

3. Download the zImage created in 2. to the Sandgate VP, and the CF card can be used to boot it if SandgateVP is started up with the CF card inserted in it. The following log is displayed during startup.

```
RedBoot(tm) bootstrap and debug environment [ROM]
Non-certified release, version Tiger.v004-004 - built 14:50:04, Oct  2 2006

Platform: PXA270 Sandgate-VP (XScale)
Copyright (C) 2000, 2001, 2002, Red Hat, Inc.

RAM: 0xa0000000-0xa4000000, 0xa000e1a8-0xa3f9d000 available
FLASH: 0x10000000 - 0x18000000, 512 blocks of 0x00040000 bytes each.
== Executing boot script in 3.000 seconds - enter ^C to abort
RedBoot> fis load zImage
RedBoot> exec -c "console=ttyS0,115200 mem=64M@0xa0000000 root=/dev/hda1 rootfst
ype=ext2"
Using base address 0xa1000000 and length 0x00200000
Uncompressing Linux.....................................................................
..... done, booting the kernel.
Linux version 2.6.17 (root@PXALinuxOsFedora5) (gcc version 4.1.0) #3 PREEMPT Mon
 Oct 2 17:59:23 JST 2006
CPU: XScale-PXA270 [69054117] revision 7 (ARMv5TE)
Machine: Sophia Systems Sandgate-VP Development Platform
Memory policy: ECC disabled, Data cache writeback
Run Mode clock: 208.00MHz (*16)
Turbo Mode clock: 416.00MHz (*2.0, active)
Memory clock: 208.00MHz (/2)
System bus clock: 208.00MHz
CPU0: D VIVT undefined 5 cache
CPU0: I cache: 32768 bytes, associativity 32, 32 byte lines, 32 sets
CPU0: D cache: 32768 bytes, associativity 32, 32 byte lines, 32 sets
Built 1 zonelists
```

```
Kernel command line: console=ttyS0,115200 mem=64M@0xa0000000 root=/dev/hda1 root
fstype=ext3
PID hash table entries: 512 (order: 9, 2048 bytes)
Console: colour dummy device 80x30
Dentry cache hash table entries: 8192 (order: 3, 32768 bytes)
Inode-cache hash table entries: 4096 (order: 2, 16384 bytes)
Memory: 64MB = 64MB total
Memory: 61824KB available (2508K code, 472K data, 96K init)
Mount-cache hash table entries: 512
CPU: Testing write buffer coherency: ok
NET: Registered protocol family 16
sandgate_vp_keypad_read_matrix: Data invalid(000000ff)
[sandgate_vp_i2c_read:875]
[sandgate_vp_i2c_read:875]
voltage: 0
NET: Registered protocol family 23
NET: Registered protocol family 2
IP route cache hash table entries: 512 (order: -1, 2048 bytes)
TCP established hash table entries: 2048 (order: 1, 8192 bytes)
TCP bind hash table entries: 1024 (order: 0, 4096 bytes)
TCP: Hash tables configured (established 2048 bind 1024)
TCP reno registered
PXA27x Dynamic Power Management
PXA27x DPM: Initial frequency is 416000kHz.
NetWinder Floating Point Emulator V0.97 (double precision)
JFFS2 version 2.2. (NAND) (C) 2001-2003 Red Hat, Inc.
Initializing Cryptographic API
io scheduler noop registered (default)
Console: switching to colour frame buffer device 30x40
SA1100 Real Time Clock driver v1.00
SA1100/PXA2xx Watchdog Timer: timer margin 60 sec
pxa2xx-uart.0: ttyS0 at MMIO 0x40200000 (irq = 21) is a BTUART
pxa2xx-uart.1: ttyS1 at MMIO 0x40100000 (irq = 22) is a FFUART
pxa2xx-uart.2: ttyS2 at MMIO 0x40700000 (irq = 20) is a STUART
loop: loaded (max 8 devices)
Uniform Multi-Platform E-IDE driver Revision: 7.00alpha2
ide: Assuming 50MHz system bus speed for PIO modes; override with idebus=xx
ide-floppy driver 0.99.newide
Probing Intel PXA27x On-Chip ROM at physical address 0x00000000 (32-bit bankwidt
h)
```

Intel PXA27x On-Chip ROM: Found 2 x16 devices at 0x0 in 32-bit bank
Intel PXA27x On-Chip ROM: Found 2 x16 devices at 0x4000000 in 32-bit bank
  Intel/Sharp Extended Query Table at 0x010A
  Intel/Sharp Extended Query Table at 0x010A
  Intel/Sharp Extended Query Table at 0x010A
  Intel/Sharp Extended Query Table at 0x010A
  Intel/Sharp Extended Query Table at 0x010A
Using buffer write method
cfi_cmdset_0001: Erase suspend on write enabled
Searching for RedBoot partition table in Intel PXA27x On-Chip ROM at offset 0x7f
c0000
5 RedBoot partitions found on MTD device Intel PXA27x On-Chip ROM
Creating 5 MTD partitions on "Intel PXA27x On-Chip ROM":
0x00000000-0x00040000 : "RedBoot"
0x00040000-0x03240000 : "jffs2"
0x03240000-0x03440000 : "zImage"
0x07f80000-0x07f81000 : "RedBoot config"
mtd: partition "RedBoot config" doesn't end on an erase block -- force read-only
0x07fc0000-0x08000000 : "FIS directory"
pxa27x_udc: version 23-Sep-2004
input: pxa27x-keypad as /class/input/input0
input: sandgatevp keypad driver!
input: pxa27x-pointer as /class/input/input1
input: sandgatevp pointer driver!
i2c /dev entries driver
enable_irq(18) unbalanced from c00183f4
I2C: Adding PXA i2c adapter.
I2C: Scanning bus (02)<5>pccard: PCMCIA card inserted into slot 0
pcmcia: registering new device pcmcia0.0
(03)<5>pccard: PCMCIA card inserted into slot 1
pcmcia: registering new device pcmcia1.0
(04)(05)(06)(07)(08)(09)(0a)(0b)(0c)(0d)(0e)(0f)(10)(11)(12)(13)(14)(15)(16)(17)
(18)(19)(1a)(1b)(1c)(1d)(1e)(1f)(20)(21)(22)(23)(24)(25)(26)(27)(28)(29)(2a)(2b)
(2c)(2d)(2e)(2f)(30)(31)(32)(33)(34)(35)(36)(37)(38)(39)(3a)(3b)(3c)(3d)(3e)(3f)
(40)(41)(42)(43)(44)(45)(46)(47)(48)(49)(4a)(4b)(4c)(4d)(4e)(4f)(50)(51)(52)(53)
(54)(55)(56)(57)(58)(59)(5a)(5b)(5c)(5d)(5e)(5f)(60)(61)(62)(63)(64)(65)(66)(67)
(68)(69)(6a)(6b)(6c)(6d)(6e)(6f)(70)(71)(72)(73)(74)(75)(76)(77)(78)(79)(7a)(7b)
(7c)(7d)(7e)(7f)
I2C: Successfully added bus
I2C: Adding PXA power i2c adapter.

```
I2C: Scanning bus (14)(3b)(50)(51)(52)(53)
PWRI2C: Successfully added bus
I2C: PXA algorithm module loaded.
ac97_codec: AC97 Audio codec, id: AKM16 (Asahi Kasei AK4650)
TCP bic registered
NET: Registered protocol family 17
hda: SanDisk SDCFB-512, CFA DISK drive
ide0 at 0xc4860000-0xc4860007,0xc486000e on irq 42
hda: max request size: 128KiB
hda: 1000944 sectors (512 MB) w/1KiB Cache, CHS=993/16/63
 hda: hda1
ide-cs: hda: Vpp = 0.0
kjournald starting.  Commit interval 5 seconds
EXT3 FS on hda1, internal journal
EXT3-fs: recovery complete.
EXT3-fs: mounted filesystem with ordered data mode.
VFS: Mounted root (ext2 filesystem). ← Mount is successful
Freeing init memory: 96K
INIT: version 2.86 booting
[STATE: 0] /etc/lids/lids.conf
[STATE: 1] /etc/lids/lids.boot.conf
        ACL Discovery is OFF
        Compiling into /etc/lids/lids.boot.acl
        Total 14 ACLs

[STATE: 2] /etc/lids/lids.postboot.conf
        ACL Discovery is OFF
        Compiling into /etc/lids/lids.postboot.acl
        Total 13 ACLs

[STATE: 3] /etc/lids/lids.shutdown.conf
        ACL Discovery is OFF
        Compiling into /etc/lids/lids.shutdown.acl
        Total 12 ACLs

INIT: Entering runlevel: 3

(none) login:
```

■ CF mount

Start up the kernel with the CF inserted.

```
//
// Kernel startup
//
#[] mkdir    /mnt/cf
#[] mount    /dev/hda1    /mnt/cf
#[] ls    /mnt/cf/
//
// The CF card contents are displayed.
//
```